

Social Robots as Behaviour-Change Intervention Agents: Implications for Human

Autonomy and Legal Accountability

CA1: Case Study

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Introduction

Generative AI is at the forefront of technological advancement, demonstrating continuous capabilities in replicating human behaviour through outputs resembling given inputs [1]. For instance, when I ask ChatGPT, one of the most popular forms of Gen AI, to "Explain psychology in three words, as if you were a 7 year old," it responded: "Feelings make thinking." [2]. Alongside this, social robotics have emerged, particularly in the form of Socially Assistive Robots (SARs). SARs are intelligent robotic systems that combine physical task capabilities with social interaction, engaging users through communication and perception as social partners [3][4].

Thus far, SARs have been successfully applied across many healthcare settings including elderly care [5], and medical settings [6], with the aim of supporting both practical and psychological needs. Considering the substantial shift towards individuals adopting more daily health-conscious habits, digital apps such as myfitnesspal and strava are experiencing annual growth in Ireland [7]. With this, SARs are more continuously applied to motivate the upkeep of daily habits, particularly to influence routine behaviours such as diet and exercise [8]. However, as this technology moves into non-clinical, widely accessible settings, significant ethical and legal challenges emerge, particularly regarding the extent to which these systems could influence human decision-making, and how responsibility is assigned when harm occurs.

This case study analyses the application of a humanoid social robot used as a behaviour-change intervention to reduce high-calorie snacking. Within this context, it identifies key ethical and legal concerns associated with autonomy and liability. The analysis then proposes targeted regulatory and design-based solutions to support safer applications of this technology.

Literature Review

SARs have been widely applied in healthcare settings, demonstrating effectiveness among individuals requiring ongoing support. In dementia care, robotic companions such as PARO have been shown to reduce agitation and improve emotional wellbeing, alongside consistent evidence of enhancing mood and social interaction [9][10][11]. Similarly, robots such as NAO have been applied in paediatric oncology, where children engage with the robot as a supportive social partner, showing increased cooperation, emotional expression, and interaction [12].

SARs are now increasingly applied beyond clinical settings. A meta-analysis of 101 studies showed their effectiveness in educational settings, functioning as teaching assistants

providing curriculum support, achieving outcomes comparable to human tutors [13]. They are also used to promote daily health behaviours through coaching and motivational support [14].

Within this context, a key strength of SARs lies in their embodied and interactive nature. Research shows that physically present robots elicit greater trust and compliance and users are more likely to follow unusual or potentially questionable requests from the robots [15]. Although the authors see this in a positive light, it raises concerns about susceptibility and limited awareness of potential implications.

A growing concern within social robotics relates to the impact of machine autonomy on human autonomy. Human autonomy refers to self-rule, involving meaningful and authentic decision-making [16], while machine autonomy refers to a system's ability to operate without real-time human input [17]. However, researchers fear this attribute of the systems may inadvertently undermine human autonomy if they subtly steer behaviour without conscious awareness, disrupting one's ability to self-determine, a core aspect of autonomy, with their predictive capabilities and nudging mechanisms [18]. Furthermore, [19] highlights that overreliance on socially assistive technologies can further weaken autonomy-related competencies, as users may become dependent on systems to structure routines and make decisions.

Closely linked to these concerns are challenges surrounding accountability and liability. Traditional frameworks assign liability clearly based on fault or product defects [20]. However, considering the socially interactive processes of the current technology, outcomes emerge from ongoing human–robot interaction rather than identifiable faults. This makes responsibility difficult to attribute, creating a “responsibility gap” [21]. Existing EU liability frameworks remain incomplete, as product-based rules do not capture the context-dependent nature of such systems, and the withdrawal of the AI Liability Directive has left further ambiguity in fault-based accountability. Consequently, responsibility remains uncertain across developers, deployers, and users [22].

Social Robots as behaviour Change Agents

[23] conducted a pilot randomised controlled trial examining the use of a humanoid social robot (NAO) as an autonomous treatment agent to deliver a behaviour change intervention targeting high-calorie snacking. Participants were recruited from the general

population based on self-reported snacking behaviours and motivation to change, and then assigned to immediate or delayed groups.

The intervention was based on Functional Imagery Training (FIT), a motivation-based technique using guided mental imagery. The robot delivered this through structured, pre-programmed, spoken prompts, guiding participants through goal-setting and motivational exercises.

Participants in the immediate group reduced snack episodes by approximately 51% within the first four weeks, with overall reductions of up to 62% observed across both groups by the end of the study. Participants also reported increased confidence in their ability to control their eating habits. Notably, the study found evidence of a working alliance between participants and the robot, suggesting that users engaged with the system in a way comparable to a human relationship.

Legal and Ethical Implications

While the findings highlight the potential of robotic systems to support nutritional behaviour change, they overlook several important ethical and legal concerns.

1. Autonomy

Firstly, the robot operated with limited capacity to adapt to user responses, yet still demonstrated how embodied systems can influence behaviour, raising concerns for user autonomy in non-clinical contexts. The perceived “working alliance” further strengthens this concern. Empirical research shows that companion-like social robots can establish therapeutic-style relationships with users, increasing trust, engagement, and willingness to continue interaction, thereby increasing influence over user behaviour [24]. In this context, the robot’s influence is not merely informational, but relational. This aligns with concerns that AI-driven systems may shape behaviour through subtle forms of nudging and personalised feedback, potentially bypassing reflective decision-making processes [25]. From an ethical perspective, this raises concerns about manipulation. Authors claim an influence is manipulative when it does not sufficiently engage a person’s capacity for reflection and deliberation [25, p23]. As NAOs structured prompts and embodied presence were sufficient to manipulate behaviour, in more advanced AI systems with adaptive learning from the user, this influence may become significantly stronger and less transparent. As a result the combination of anthropomorphism, trust, and behavioural prompting may undermine fully autonomous decision-making, particularly for those with impaired self-regulation.

2. Liability and Accountability

The responsibility gap identified in the literature becomes evident within this specific case. Under the EU AI Act, Article 16 assigns responsibility to providers to ensure systems are safe and compliant [26]. Articles 13 and 14 then introduce the expectation of user involvement [27, 28]. However, this does not clearly transfer liability away from providers, instead creating a shared and somewhat ambiguous responsibility.

The EU Medical Device Regulation provides clearer liability pathways. With AI systems that are classified as medical devices, responsibility falls on manufacturers for defective or unsafe products [29]. However, in this case, NAO, when used as a behaviour-change tool to reduce snacking, would not likely qualify as a medical device, so we turn to Article 56, which introduces codes of practice for general-purpose AI [30]. Unfortunately, further limitations arise as these rely on voluntary compliance rather than strict enforcement, and are therefore insufficient to guarantee user protection. Taken together, in non-clinical contexts, responsibility becomes highly uncertain.

This uncertainty is particularly problematic where harm emerges gradually through behavioural influence rather than a single identifiable failure. If NAO contributes to maladaptive eating behaviours, potentially escalating into an eating disorder, it is unclear where liability lies. This concern is not purely theoretical, as similar risks have been identified in digital health tools, for example, 75% of individuals with an eating disorder reported using MyFitnessPal, with 73% stating that it contributed to their condition [31].

Ultimately, this case exemplifies the severe regulatory gap. While the proposed EU AI Liability Directive offered some prospect of addressing these challenges, its withdrawal has significantly exacerbated the issue. As it stands, neither provider-focused obligations nor user-focused requirements provide a definitive basis for liability, leaving behaviour-influencing AI systems insufficiently governed.

Proposed Solutions

In light of these concerns, targeted regulatory and design-based interventions are required to ensure a safer and more accountable robotically driven environment.

Firstly, to protect user autonomy, social robots should be designed to prompt active decision-making. For example, systems could incorporate reflective prompts, for example,

‘Are you sure?’, encouraging users to engage their deliberative capacities [32]. This approach preserves authentic choice and reflective agency, both of which are central to autonomy [19]. In addition, excessive anthropomorphic design, particularly emotional cues that foster dependency [33] or a perceived “working alliance” [23], should be limited, as they may distort user preferences and compromise authentic decision-making [34].

Furthermore, the legal frameworks surrounding behaviour-change AI should be updated to require explicit responsibility assignment at every stage of deployment, from development to deployment, and during user interaction. For instance, developers must evaluate the potential for maladaptive outcomes, such as disordered eating, before deployment, and deployers should ensure that the system is properly implemented and that users are informed about its capabilities and risks. By identifying clear roles at each stage, this solution eliminates ambiguity about who is responsible for what.

To strengthen this, this liability pathway should be strictly enforced. The voluntary nature of Article 56 Codes of Practice needs to be replaced with binding obligations, supported by mechanisms and penalties. This would reduce reliance on industry cooperation and establish clearer lines of responsibility.

Conclusion

As socially assistive robots move beyond clinical settings into everyday behaviour change, significant ethical and legal concerns arise alongside them. This case study has shown that while such systems can be effective, they risk undermining autonomy through behavioural influence, while also creating uncertainty in legal accountability. Existing frameworks remain insufficient, leaving behaviour-influencing systems operating within a regulatory gap. Addressing this requires stronger safeguards, clearer responsibility, and a regulatory reform that reflects their behavioural impact.

Please note that OpenAI was used in the preparation of this case study for structure and grammar, strictly in accordance with DCU’s guidelines for AI use [2].

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